

# Aditya Rao

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## EDUCATION

### McMaster University

*B.Eng in Software Engineering (CO-OP)*

Hamilton, ON

Sept 2023 – Apr 2028

- 4.00/4.00 CGPA - Dean's Honour List
- Relevant Coursework: Digital Systems, OOP, Discrete Math I & II, Intro to Software Design, DSA, Computer Architecture, Software Development Basics

## EXPERIENCE

### Software Developer Intern

May 2025 – Aug 2026

IBM

Markham, ON

- Developing WebSphere Application Server v8.5.5, v9.0, WebSphere Liberty, and OpenLiberty and the public and internal tools which support development.
- Re-engineered a suite of 4 internal legacy tools from scratch, modernizing the UX, standardizing Docker deployments, and establishing the first Unit/Integration testing pipelines to eliminate manual verification.
- Created a tool used by over 400+ developers and managers which allows developers to record the distribution of their work they perform and allow managers to record it and use the data to assist the developers by allocating resources in their workflows.
- Optimized the core data processing tool within this suite, replacing heavy framework overhead with custom API wrappers to reduce execution time by 99% (from 10 mins to <4 secs).
- Refactored the critical Product Installation Verification Suite, migrating 20+ legacy FAT tests to a unified framework which enhanced debuggability and system reliability.
- Diagnosed and solved >50 build breaks and supported the development team in diagnosing customer cases and build breaks on all platforms, including but not limited to Linux, Windows, MacOS, IBM i, and zOS
- Created E2E tests for deploying Liberty on OCP (Kubernetes) and verifying behavior along with using third-party software such as Infinispan, Keycloak, and Kafka.

### Full Stack Developer Intern

May 2024 – Aug 2024

Medium AI

Hamilton, ON

- Developed and maintained React and TypeScript code for Medium AI, contributing to 2 first-place and 1 second-place finishes in the McMaster Startup Survivor competition, and helping secure \$10,000 in funding.
- Implemented real-time speech transcription using Python, OpenAI Whisper, FFmpeg, and Socket.IO

## PROJECTS

### WebCraft | 🌐 | C++, CMake, Vcpkg, Asynchronous Programming, Github Actions

Apr 2025 – Sept 2025

- Developed an asynchronous runtime and programming library using C++ 23 features to allow developers who are used to Node.js, Python, and Java to write fast and simple async C++ code easily.
- Created coroutine abstractions, the async runtime, async streams, file and networking module abstractions over low level OS specific async library code to simplify cross-platform C++ development.
- Created CI/CD pipelines and created over 130+ tests to verify correctness using gtest, CTest, Vcpkg, and Github Actions to ensure code quality and reliability.
- Created a vcpkg port so developers can easily access it.

### Resume Builder | 🌐 | Java, Spring Boot, Next.js, Typescript, Python, FastAPI, LangChain, Docker

Aug 2024 – Feb 2025

- Developed an AI-powered resume builder for automated resume creation, scoring, and real-time editing.
- Improved resume management by reducing manual effort in tailoring resumes for job applications.

### FashioNova | 🌐 🌐 | Typescript, Next.js, Python, Socket.io, OpenCV, Cloudflare AI, Flask

Oct 2024 – Oct 2024

- Developed a virtual wardrobe and AI fashion advisor in 36 hours at Hack the Valley 9 at UofT Scarborough
- Created an OpenCV pipeline using Python Flask and Socket.io to fit virtual clothes onto users via live video feed
- Leveraged Cloudflare AI to suggest outfits based on the occasion, promoting sustainable fashion consumption.

### Smart Study Buddy | 🌐 🌐 | Python, Flask, Wolfram Alpha API, OpenAI API, Next.js, Typescript

Oct 2024 – Oct 2024

- Developed a virtual wardrobe and AI fashion advisor in 36 hours at Hack the Valley 9 at UofT Scarborough
- Created an OpenCV pipeline using Python Flask and Socket.io to fit virtual clothes onto users via live video feed
- Leveraged Cloudflare AI to suggest outfits based on the occasion, promoting sustainable fashion consumption.

### QuakeGuard | 🌐 🌐 | Next.js, Flask, Typescript, Python, TensorFlow, Keras, SciKit Learn, Pandas

Jan 2024 – Jan 2024

- Developed an earthquake prediction tool using ML models showcased at DeltaHacks X and GDSC McMaster
- Visualized global earthquake data and high-risk areas with Next.js and Python, Flask, SciKit Learn, and TensorFlow

EXTRA CURRICULARS

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| <b>Sys-Admin and Fullstack Developer Executive</b><br><i>IEEE McMaster Student Branch</i> <ul style="list-style-type: none"><li>• Set up an RPi 3B+ Cluster for a server to handle up to 100 clients in an instant with a budget of \$5000</li><li>• Managed and provisioned a CI/CD pipeline which generates Docker images for our Kubernetes cluster</li><li>• Developing the website for the IEEE student branch and the admin dashboard for executives to manage</li></ul> | Nov 2024 – Present<br><i>Hamilton, ON</i> |
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TECHNICAL SKILLS

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| <b>Languages:</b> Java, Python, SQL, .NET Framework, C, C++, JavaScript/TypeScript, Verilog<br><b>Frameworks:</b> Spring Boot, Jakarta EE, MicroProfile, Hibernate, Express, Next.js, FastAPI, Flask, UWP<br><b>Developer Tools:</b> VS Code, IBM Bob, Eclipse, AWS, GCP, Git, Github, Docker, OpenShift, Kubernetes, Databases, Infinispan<br><b>Libraries:</b> Pandas, TensorFlow, OpenCV, SciKit Learn, JavaFX, Redux, JQuery, React, Mongoose<br><b>Hobbies:</b> Sketching, Painting, Violin (Western and Carnatic), Badminton, Swimming, Chess |  |
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CERTIFICATIONS

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| <b>Clean Code</b><br><i>Udemy</i>                                                   | September 2025 |
| <b>IBM Agile Explorer</b><br><i>IBM</i>                                             | September 2025 |
| <b>The DevOps Essentials - The Handbook</b><br><i>Udemy</i>                         | September 2025 |
| <b>Introduction to Containers, Kubernetes, and OpenShift</b><br><i>IBM</i>          | August 2025    |
| <b>Practical OpenShift for Developers - OpenShift 4</b><br><i>Udemy</i>             | August 2025    |
| <b>[NEW] Ultimate AWS Certified Cloud Practitioner CLF-C02 2025</b><br><i>Udemy</i> | February 2025  |
| <b>Docker &amp; Kubernetes: The Practical Guide [2025 Edition]</b><br><i>Udemy</i>  | January 2025   |